Structure-based functional annotation of hypothetical proteins using the AlphaFold Database and TM-align

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Structure-based functional annotation of hypothetical proteins using the AlphaFold Database and TM-align: a laptop-scale, reproducible pipeline

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Abstract

Background. Every sequenced genome contains proteins annotated only as "uncharacterized", "hypothetical", or "domain of unknown function": their amino-acid sequence matched no characterised protein, so no biological function was assigned. Because three-dimensional structure is conserved far longer than sequence, a protein can closely resemble a well-studied enzyme in shape while being undetectable by sequence search. The public release of >200 million predicted structures in the AlphaFold Protein Structure Database (AlphaFold DB) makes it possible, in principle, to re-examine these proteins at the level of fold rather than sequence.

Results. We present *Protein Function Rescue*, an open, reproducible pipeline that (i) retrieves an organism's AlphaFold models, (ii) filters them by the AlphaFold per-residue confidence (pLDDT), (iii) identifies the currently unannotated proteins from their UniProt records, (iv) searches each against a reference library of functionally characterised enzymes using TM-align, (v) characterises the best hits' putative binding cavities with a geometric (LIGSITE-style) pocket detector, and (vi) ranks the resulting candidate function assignments with a transparent composite score. The entire workflow runs on a commodity laptop with no GPU and no compiled dependencies, and ships with a static web viewer for inspecting each candidate in 3D. In a pilot on nine unannotated *Mycobacterium tuberculosis* proteins searched against a 119-protein reference library, four proteins matched a known fold or domain at a query-normalised TM-score ≥ 0.5. Three larger proteins yielded fold-level hypotheses; the cleanest whole-chain match was the uncharacterized P9WQ67 against an aromatic amino-acid aminotransferase (query/reference-normalised TM-scores 0.67/0.63), although catalytic-residue identity was not conserved and reaction transfer remains unproven. A 50-residue fourth hit was identified as a short-fragment false positive. On an independent benchmark of 227 enzymes across 29 families and six EC classes, leave-one-out recovery assigned the exact four-level EC number correctly in 96.5 % of cases, versus 93.8 % for a deliberately simple symmetric pairwise-identity baseline at the exact-reaction level. No paired significance analysis was performed. Twenty-four of 219 correct structural calls had <30 % identity to their structural hit; these are low-identity recoveries, not necessarily 24 cases where structure defeated the sequence baseline. At the default TM-score threshold of 0.5, precision was 0.982.

Conclusions. Structure-based triage recovers specific, testable functional hypotheses for proteins that sequence-based annotation leaves blank, and can be performed at classroom/laptop scale. The method is presented as a hypothesis-generating tool; all predictions require experimental validation.

Keywords: AlphaFold; structural bioinformatics; protein function prediction; hypothetical proteins; TM-align; binding-pocket detection; *Mycobacterium tuberculosis*; remote homology.

1. Introduction

The gap between the number of known protein *sequences* and the number of proteins with an experimentally established *function* continues to widen. A large fraction of every proteome — often 25–50 % — consists of proteins with no assigned function, catalogued under labels such as “hypothetical protein”, “uncharacterized protein”, or “domain of unknown function (DUF)” [1]. This “dark proteome” is not biological noise: it includes enzymes, transporters, and regulators of real physiological importance, and in pathogens it may contain untapped drug or vaccine targets [1].

Historically, function has been transferred between proteins by *sequence* homology. This approach fails precisely where it is most needed: once two proteins have diverged past roughly 20–25 % sequence identity, homology becomes difficult to detect from sequence alone, even though the proteins may retain the same fold and function. Structure is empirically three to ten times more conserved than sequence [2], so structural comparison can recover relationships that sequence comparison misses.

Two developments now make structure-based annotation feasible at scale. First, AlphaFold [3] achieved near-experimental accuracy in single-chain structure prediction, and the AlphaFold Protein Structure Database (AlphaFold DB) [4] released predicted structures for essentially the entire UniProt reference proteome set — over 200 million models — under an open licence, each accompanied by a per-residue confidence estimate (pLDDT). Second, fast structural aligners and search tools such as Foldseek [5], built on the TM-align algorithm and TM-score metric [6, 7], make all-versus-all structural comparison computationally tractable.

Beyond direct structural search, machine-learning methods now infer function directly from structure: DeepFRI [12], for example, predicts Gene Ontology terms and EC numbers from a graph-convolutional encoding of the fold. Such learned predictors and fast search tools provide complementary capabilities, often through local installations, pre-indexed resources, or hosted web services. Recent reviews and community assessments of AlphaFold’s impact on function prediction [13, 14] make two points that frame the present work: the opportunity created by 200 million structures is enormous, yet structure alone is not sufficient — predictions remain hypotheses that require experimental validation, and the field is still evolving rather than solved.

Positioning and research question. Existing work has established that AlphaFold structures and structural search can improve annotation. We ask a narrower implementation question: *can a transparent, Windows-compatible teaching workflow achieve meaningful annotation performance on commodity hardware while remaining fully* *reproducible?* Concretely, we target a single laptop with no GPU, no administrator privileges, and no compiled bioinformatics binaries, assembling the entire workflow from the pre-computed AlphaFold DB plus pure-Python components, and we apply it to unannotated proteins of *Mycobacterium tuberculosis* (Mtb), a World Health Organization priority pathogen whose genome remains rich in conserved hypothetical proteins.

Our contribution is fourfold: (1) an explicit demonstration of fold-level triage on commodity hardware, without a local GPU or compiled command-line bioinformatics executable; (2) a transparent, end-to-end pipeline that turns raw AlphaFold models into ranked, evidence-linked functional hypotheses, in which the compute-heavy stages (structural search, pocket detection) are replaced by laptop-friendly, fully interpretable equivalents; (3) a controlled validation quantifying recovery accuracy against a sequence baseline, with residue-level active-site verification and ranking-robustness analysis; and (4) an openly available, reproducible software tool with an interactive viewer. Our approach is deliberately **complementary** to the prior art: unlike learned predictors such as DeepFRI [12], each assignment is traceable to a specific structural match and a coarse residue-site check; and compared with Foldseek [5] it trades database scale and speed for a small, inspectable reference set.

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2. Methods

2.1 Overview

The pipeline comprises six stages executed in sequence: **download → parse → annotate → search → pocket → rank** (Figure 1). Each stage caches its output to disk, so any stage can be re-run independently. The structural-search and pocket-detection stages each expose two interchangeable back-ends — a native, pure-Python implementation (default) and an external high-performance tool — so the identical workflow runs on a laptop or scales up on a Linux server.

2.2 Data sources and structure retrieval

Predicted structures were obtained from AlphaFold DB [4] (model version 6). For the target organism, *M. tuberculosis* H37Rv (UniProt reference proteome UP000001584, NCBI taxonomy 83332), individual models were retrieved as mmCIF files; per-structure URLs were resolved through the AlphaFold DB API to remain robust to database version changes. For whole-proteome runs the pipeline instead downloads the organism's proteome archive from the AlphaFold DB FTP site. Structures were parsed with the *gemmi* library [8].

2.3 Confidence filtering

AlphaFold reports a per-residue confidence, pLDDT (0–100), stored in the B-factor field of each model [3]. For each structure we computed the mean pLDDT over Cα atoms and discarded models below a configurable threshold (default 70), on the rationale that low-confidence models yield unreliable structural comparisons. Residue-level pLDDT is retained and used to colour structures in the viewer so that low-confidence regions are visually flagged.

2.4 Annotation status

Each protein's current annotation was retrieved from UniProt [9] (protein name, evidence level, and keywords). A protein was flagged as *unannotated* — and therefore a candidate for structural rescue — if its recommended name contained any of the markers `uncharacterized`, `hypothetical`, `DUF`, or `putative uncharacterized`, or if its UniProt protein-existence evidence was "Predicted" or "Uncertain". Only flagged proteins were carried forward as queries.

2.5 Structural search (TM-align back-end)

Function was transferred by structural similarity to a reference library of *functionally characterised* proteins. The reference library was assembled automatically by querying UniProt for reviewed (Swiss-Prot) entries carrying an Enzyme Commission (EC) number and protein-level existence evidence (`(reviewed:true) AND (ec:*) AND (existence:1) AND (fragment:false)`), ranked by annotation score; their AlphaFold models were then downloaded. For the pilot, 120 references were requested and 119 were successfully retrieved.

Each query structure was aligned to every reference using TM-align [6] via the *tmtools* Python bindings, operating on Cα coordinates and sequence extracted with *gemmi*. TM-align returns a TM-score [7], a length-normalised measure of fold similarity in the range 0–1, where values > 0.5 generally indicate the same fold and values near 0.2 correspond to unrelated structures. Because TM-score can be normalised by the length of either chain, we retained the larger value as a domain-sensitive screening score but recorded both normalisations, both lengths, and RMSD. A high query-normalised but low reference-normalised score indicates a partial/domain match and is not evidence that the complete proteins share a function. For each query, the reference with the highest screening score was retained if it met a configurable threshold (default 0.5).

TM-align performs an exhaustive, exact pairwise alignment; it is therefore slower than heuristic tools but requires no database construction and no compiled dependencies. As an alternative back-end, the pipeline can instead call Foldseek [5] against a pre-built PDB or Swiss-Prot database, which is far faster and scales to millions of targets; the two back-ends are selected by a single configuration key.

2.6 Binding-pocket detection (geometric back-end)

For each retained hit we characterised the largest putative binding cavity using a compact, LIGSITE-style geometric algorithm [10, 11] implemented in NumPy/SciPy:

1. The protein was rasterised onto a 3D grid (default spacing 1.2 Å, 5 Å padding); grid points within the van der Waals radius of any heavy atom were marked "protein".

2. For each empty (solvent) grid point, we counted, over seven scan directions (the three axes plus four body diagonals), how many directions had protein on *both* sides — a protein–solvent–protein (PSP) "sandwich" event, indicating enclosure.
3. Solvent points enclosed in at least five of seven directions were labelled pocket points.
4. Adjacent pocket points were clustered (connected-component labelling); clusters below a minimum volume were discarded. The number of surviving clusters gave the pocket count.

For the largest pocket we computed a geometric *cavity score* in [0, 1] by combining its volume (via a logistic transform) with its mean enclosure (buriedness). We emphasise that this cavity score is an unsupervised geometric estimate, **not** the trained druggability score produced by fpocket [11]; it is labelled as such throughout the software and in the results. As with structural search, fpocket is available as an alternative back-end for users on Linux/macOS.

2.7 Composite ranking

Each candidate received a composite score combining three normalised signals: structural similarity (TM-score), model confidence (mean pLDDT / 100), and pocket quality (cavity score). With default weights $w = (0.5, 0.2, 0.3)$, the composite is the weighted mean over the available components; if a component is missing (e.g. no pocket detected), its weight is dropped and the remaining weights are renormalised:

$$\text{composite} = \Sigma_i w_i v_i / \Sigma_i w_i \text{, over available components } i.$$

Candidates were sorted by composite score in descending order.

2.8 Implementation, software, and availability

The pipeline is written in Python 3.12 and depends only on packages installable with `pip` (*requests*, *gemmi*, *pyyaml*, *pandas*, *tqdm*, *numpy*, *scipy*, *tmtools*, *biopython*, *matplotlib*). Results are exported as JSON and CSV, and as a JavaScript data file consumed by a static, dependency-free web viewer that renders each structure in the browser (3Dmol.js), coloured by pLDDT, alongside its structural match, pocket statistics, and composite score. No server, database, or GPU is required. The software, configuration, and pilot outputs are released under the MIT licence; AlphaFold data are used under the EMBL-EBI terms and are for theoretical modelling only.

2.9 Functional-residue verification

Global fold similarity is necessary but not sufficient evidence for shared function, so we added an optional residue-level verification step. For a candidate–reference match we retrieved the reference's annotated functional residues from UniProt (active-site, binding-site and metal-binding features), superposed the candidate onto the reference with the TM-align rotation/translation, and followed TM-align's explicit residue correspondence. A reference functional residue was scored as *position-compatible* if its aligned candidate Cα lay within 4 Å after superposition, and additionally *identity-conserved* if that candidate residue was the same amino acid. The fractions of position-compatible, identical, and chemically conservative aligned sites provide a coarse check that complements the global score; they do not establish side-chain geometry or catalytic equivalence (implemented in `pipeline/functional_residues.py`).

2.10 Weight-sensitivity analysis

To test whether candidate rankings depend on the specific composite weights, we scored a pool of proteins and re-ranked them under every weight vector on the three-component simplex (each weight ≥ 0.05, in 0.05 steps), measuring the Kendall rank correlation and top-10 overlap of each ranking against the default-weight ranking (`benchmark/weight_sensitivity.py`).

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3. Results

3.1 Pilot dataset

As a proof of concept we processed a pilot set of ten *M. tuberculosis* proteins annotated by UniProt as uncharacterized/hypothetical. One model fell below the mean-pLDDT threshold (70) and was discarded, leaving nine high-confidence queries (mean pLDDT range 75.7–97.1). These were searched against the 119-protein reference library of characterised enzymes (Section 2.5). This pilot is intentionally small and is meant to demonstrate the method and software end-to-end; whole-proteome application is straightforward but was outside the scope of this demonstration (see Limitations).

3.2 Candidate function assignments

Four of the nine queries (44 %) produced a domain-sensitive screening hit at TM-score ≥ 0.5 (Table 1). Because this threshold uses the larger length normalisation, these are fold/domain matches rather than automatic function assignments. The remaining five had no reference above threshold.

Table 1. Ranked structural-screening hits for the pilot set. TM-score Q/R gives query- and reference-length normalisations; pLDDT is mean AlphaFold confidence; the cavity score is the native geometric estimate. Matches are hypotheses, not experimental assignments.

Rank	Query	Length Q/R	Mean pLDDT	Best structural match	TM-score Q/R	RMSD (Å)	Cavity score	Interpretation
1	P9WI	459/1022	95.5	D-2-hydroxyglutarat	0.904/0.421	2.647	0.966	Oxidoreductase-

Rank	Query	Length Q/R	Mean pLDDT	Best structural match	TM-score Q/R	RMSD (Å)	Cavity score	Interpretation
	T1			e dehydrogenase				related domain; specific reaction unsupported
2	P9WJG7	50/269	80.0	TIR-domain NAD$^+$ hydrolase	0.862/0.180	1.138	—	Short-fragment false positive
3	O08343	264/480	97.1	Ochratoxinase / amidohydrolase 2	0.715/0.423	3.461	0.968	Metallo-hydrolase architecture; substrate specificity unsupported
4	P9WQ67	398/423	96.4	Aromatic amino-acid aminotransferase	0.666/0.633	4.014	0.968	Aminotransferase-fold hypothesis; catalytic identity unsupported

3.3 Interpretation of the structural hits

- **P9WIT1** is already annotated broadly as an FAD-linked oxidoreductase (EC 1.-.-.-). Its match is compatible with an oxidoreductase-related domain, but the reference is more than twice as long and the reference-normalised score is 0.421. The evidence does not refine P9WIT1 to D-2-hydroxyglutarate dehydrogenase.
- **O08343** is already a TatD-family metal-dependent hydrolase. The match and metal-site-compatible residues support that broad architecture, but do not establish ochratoxinase, peptidase, or amidase substrate specificity.
- **P9WQ67 (Rv3778c)** provides the cleanest whole-chain fold match because both TM-score normalisations exceed 0.5. It is therefore an aminotransferase-fold hypothesis, pending direct verification of PLP-binding and catalytic residues.

- **P9WJG7** is only 50 residues. Its query-normalised score is 0.862 but its reference-normalised score is 0.180, and no functional sites align. It is a false positive caused by permissive short-fragment normalisation and is retained only as a documented failure mode.

3.4 Pocket analysis

The geometric detector identified enclosed cavities in the three larger high-scoring proteins (4–6 pockets; cavity scores 0.966–0.968), consistent with globular enzymes possessing defined active sites, and reported no enclosed pocket for the 50-residue membrane peptide P9WJG7. Because the cavity score is an unsupervised geometric measure, we use it only as a supporting signal in the composite ranking and for visual inspection, not as a standalone druggability claim.

3.5 Software and interface

The pipeline produced machine-readable outputs (JSON, CSV) and a static web interface that lists the ranked candidates and renders each AlphaFold model in 3D, coloured by pLDDT, with its structural match, match confidence, pocket statistics, and per-component scores (Figure 2). The complete pilot — download, parsing, annotation, 9 × 119 TM-align comparisons, pocket detection, and ranking — ran on a consumer laptop without a GPU; the structural-search stage took 18 min 37 s in the verification run. Runtime was strongly length-dependent, with the 1,327-residue P9WN15 query dominating the run.

3.6 Functional-residue verification of the candidates

Global fold similarity alone leaves open whether the *active site* is shared. We therefore tested, for each pilot candidate, whether the matched reference's annotated catalytic and binding residues are compatible under the explicit TM-align correspondence (Section 2.9; Table 2).

O08343 is the only strong residue-level case: all eight annotated reference sites have aligned Cα pairs within 4 Å, and four identities are retained. Reference H111/H113/D378 correspond to candidate H5/H7/D208; those candidate sites are already annotated by UniProt by similarity, so this supports metal-site compatibility rather than discovering a new reaction. **P9WIT1** has no aligned functional site within 4 Å; two aligned identities lie outside the distance cutoff and the reference Fe-S cysteines fall outside the matched region. **P9WQ67** has 9/11 position-compatible sites but no identities or conservative substitutions, so its catalytic machinery is unverified. **P9WJG7** has no aligned functional sites. The check is a coarse Cα proxy and does not test side-chain coordination geometry.

Table 2. Functional-site compatibility for the pilot hits. Sites use TM-align's residue correspondence; position-compatible means the aligned Cα pair is within 4 Å. Identity is counted only for position-compatible pairs.

Candidate	Matched enzyme	Sites	Position-compatible	Identity-conserved	Interpretation
O08343	Amidohydrolase (Zn)	8	8 (100 %)	4 (50 %)	Candidate H5/H7/D208 support metal-site compatibility
P9WQ67	PLP aminotransferase	11	9 (82 %)	0	Fold hypothesis; catalytic identity unsupported
P9WIT1	D-2-hydroxyglutarate dehydrogenase	6	0	0	Reference functional sites not supported by aligned geometry
P9WJG7	TIR-domain NAD^+ hydrolase	6	0	0	Short-fragment false positive

4. Validation: a leave-one-out benchmark

The pilot of Section 3 shows the software produces coherent hypotheses, but coherence is not accuracy. To quantify how often structure-based recovery is *correct*, and whether it adds value over the sequence-based methods it is meant to complement, we ran a controlled leave-one-out benchmark on a labelled gold-standard set.

4.1 Benchmark design

We assembled 227 reviewed enzymes with experimentally supported Enzyme Commission (EC) annotations, drawn from 29 EC families spanning six enzyme classes (oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases; classes 1–6), with up to eight diverse members per family — families returning fewer than three usable members were dropped automatically — and lengths of 80–520 residues. For every protein we performed leave-one-out prediction: its annotation was hidden and its function was predicted from the EC number of its single best hit among all *other* proteins, computed two ways — (i) by TM-align structural similarity (the pipeline's method) and (ii) by a sequence baseline (the mean identity from local BLOSUM62 alignments in both sequence orders, making the pairwise score symmetric). Predictions were scored by EC agreement at each level (1 = class … 4 = exact reaction). This isolates the effect of the comparison method, since both operate on the identical set. Full code and the exact accession list are in `benchmark/`.

4.2 Recovery accuracy and comparison to sequence

Structure-based recovery assigned the **exact four-level EC number** correctly for **219/227 proteins (96.5 %)**, and was stable across EC levels 1–4 (Figure 3). The simple pairwise sequence-identity baseline was identical at the coarsest level (class, level 1: 96.5 %) but lower at finer levels, falling to **93.8 % for the exact reaction** (level 4). This is a 2.7-percentage-point observed difference; no paired confidence interval or significance test was performed, so we do not claim statistical superiority or equivalence. A stronger sequence method (profile/HMM search; Section 6) could narrow or reverse the difference.

4.3 Correct structural calls at low pairwise identity

Among the 219 correct structural recoveries, **24 (11 %) had below 30 % sequence identity to the structural hit** (Figure 4), including several examples at very low identity. This establishes that the structural nearest neighbour can recover the labelled EC family despite little pairwise identity to that neighbour. It does not mean that all 24 are unique wins over the sequence baseline, whose nearest hit may be a different protein. Accordingly, Figure 4 is interpreted as a low-identity structural-call analysis rather than a general comparison with modern sequence methods.

4.4 Calibration of the TM-score threshold

Precision as a function of the TM-score acceptance threshold (Figure 5) supports the default cutoff of 0.5: at TM ≥ 0.5, precision for ≥3-level EC agreement was **0.982 (219 of 223 accepted hits correct)** at 98.2 % coverage; raising the threshold to 0.75 gave **100 % precision** at 95.2 % coverage. The four false positives at the default threshold are instructive rather than random — each is a genuine case of shared fold architecture without shared function: a mutual confusion between a 94-residue pyrimidine/purine nucleoside phosphorylase and a glucose-6-phosphate isomerase (TM 0.74), a PLP-fold alanine racemase matched to an aspartate aminotransferase, and a microsomal (MAPEG-fold) prostaglandin E synthase. High structural scores reflect fold identity, which usually but not always implies functional identity.

4.5 Instructive failure modes

Of the eight proteins not correctly recovered, **four had a best structural score below the 0.5 threshold and were therefore correctly rejected rather than misassigned**; the other four are the false positives above. The rejected cases are all atypical members whose fold differs from the rest of their EC family — a carboxysomal carbonic anhydrase (CsoSCA), which adopts a fold distinct from the α-carbonic anhydrases; a lanthionine-synthetase-like protein (LANCL1) annotated as a glutathione transferase but not sharing the GST fold; and proteins carrying secondary EC labels on a different scaffold (e.g. a peptidoglycan-editing factor also annotated as a purine nucleoside phosphorylase). This behaviour is desirable:

the confidence threshold converts many fold-degeneracy cases into abstentions rather than wrong high-confidence calls, exactly as the pipeline's filtering and ranking intend.

4.6 Robustness of the composite ranking to the weights

The composite score combines structural similarity, model confidence and pocket quality with default weights (0.5, 0.2, 0.3); a reviewer will reasonably ask whether the ranking depends on those particular numbers. To test this we scored a realistic pool of 40 *M. tuberculosis* rescue candidates (structural- similarity scores spanning 0.45–0.84) and re-ranked them under many weight vectors (Section 2.10; Figure 6). Within a reasonable neighbourhood of the default — structure kept dominant, each weight varied by up to ±0.1 — the ranking was essentially unchanged: Kendall's τ against the default ranking had a **median of 0.98 (minimum 0.95; all 61 weightings above 0.9)**, and the top-10 candidates were identical for the majority of weightings (median top-10 Jaccard 1.0). Even under an adversarial stress test over the *entire* weight simplex (171 vectors, including weightings that all but ignore structure), the median τ remained 0.91. Candidate rankings are therefore driven by the data, not by the specific weights: the weights tune emphasis, not outcome. (Consistent with this, the geometric cavity score has limited dynamic range across well-folded proteins — Section 6 — so most of the ranking signal comes from structural similarity and model confidence regardless.)

4.7 Summary

On a labelled benchmark of 227 enzymes across 29 families and six classes the pipeline recovers exact enzyme function in 96.5 % of leave-one-out tests, versus 93.8 % for the simple pairwise-identity baseline at the exact-reaction level. The observed difference was not tested for statistical significance. Twenty-four correct structural calls had low identity to their structural neighbour, while the calibrated threshold keeps precision high (0.982 at TM ≥ 0.5) and its failures are dominated by the well-understood fold-degeneracy of the EC system, most of which the threshold turns into abstentions rather than errors. These are quantitative results the pilot alone could not provide; we emphasise that the sequence comparator here is a deliberately simple pairwise baseline, not a profile/HMM method.

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5. Discussion

This work demonstrates that the two ingredients underlying modern structural bioinformatics — a comprehensive database of predicted structures and a reliable structural aligner — are individually lightweight enough to be combined into a functioning function-annotation pipeline on a personal computer. By consuming pre-computed AlphaFold models rather than predicting structures, the compute cost collapses to that of pairwise structural alignment, which TM-align performs exactly and without any database-building step.

The pilot results are useful in a specific, limited sense: P9WIT1 and O08343 produced domain/fold matches coherent with their existing broad enzyme-class annotations, while P9WQ67 produced an aminotransferase-fold hypothesis where the current record contains no enzyme class. None of these results establishes a specific reaction, and the P9WJG7 result documents a short-fragment false positive.

The design also makes the speed/scale trade-off explicit and pedagogically useful. TM-align's exhaustive, exact alignment is ideal for a curated reference set on a laptop but does not scale to searching millions of targets; Foldseek's *k*-mer-based structural search was engineered precisely for that regime [5]. Because both are exposed as interchangeable back-ends, a user can prototype on a laptop and later scale the identical analysis to a whole-proteome, whole-PDB search on a Linux server by changing one configuration value, although the two search methods are not scientifically equivalent. The pocket stage mirrors this: a transparent geometric detector for laptop use, or the trained fpocket tool [11] where available.

Compared with sequence-based annotation transfer, the structural approach's advantage is its reach into the "twilight zone" of low sequence identity [2]; its disadvantage is that fold similarity does not guarantee identical function (convergent folds and moonlighting proteins exist), which is why every output is framed as a hypothesis linked to its supporting evidence rather than a definitive assignment.

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6. Limitations

Several limitations bound the interpretation of these results:

1. **Predictions are hypotheses.** A structural match proposes a possible function; it is not experimental proof. AlphaFold models are themselves predictions, and low-pLDDT regions are unreliable. None of the *M. tuberculosis* assignments has been experimentally validated. AlphaFold data are explicitly not for clinical use.
2. **Scope of the benchmark.** Validation (Section 4) was performed on 227 enzymes across 29 EC families and six enzyme classes; while a few hundred proteins is a reasonable scale, accuracy on a curated enzyme set is still an optimistic estimate of performance on the harder, more heterogeneous dark proteome. The length cap (80–520 residues, imposed to keep exhaustive TM-align tractable) also excludes the largest enzymes. The benchmark measures EC-number recovery, a proxy for function, and does not capture non-enzymatic function, multi-domain proteins, or moonlighting activities. Larger benchmarks — e.g. the full SCOPe/CATH superfamily hierarchy, or held-out proteins characterised after a fixed AlphaFold DB release — and a whole-proteome application remain future work.
3. **Comparison baseline.** We compared against a local sequence-identity baseline, the most accessible sequence method; a comprehensive evaluation should also compare against profile/HMM search (HMMER/InterProScan), the Foldseek back-

end, and learned function predictors (e.g. DeepFRI), which was beyond the scope of this laptop-based study. Some proteins carry multiple legitimate EC numbers, whereas the benchmark assigns the single family EC under which each entry was fetched; apparent errors can therefore include valid secondary functions.

4. **Reference-library bias.** Coverage and ranking depend on which characterised proteins are in the reference set; a small or skewed library will miss functions it does not contain.
5. **Length normalisation and partial matches.** Ranking by the larger TM-score normalisation is deliberately permissive for domain matches and can inflate short-to-long comparisons (illustrated by the 50-residue P9WJG7). Both normalisations, both lengths, and coverage must be inspected before function transfer.
6. **Geometric cavity score.** The pocket score is an unsupervised geometric estimate, not a validated druggability prediction, and should not be interpreted as the latter. It also has limited dynamic range across well-folded proteins (most globular enzymes score similarly), so it contributes only weakly to the composite ranking; improving pocket scoring (or substituting fpocket) is future work.
7. **Fold ≠ function.** Structural similarity can arise without functional identity, as the benchmark's TIM-barrel false positive shows. The functional-residue verification of Sections 2.9/3.6 mitigates this by checking catalytic and metal/ligand-binding residues, and is provided as a standard step; however, we have not yet benchmarked its accuracy systematically, and aligned Cα proximity is a coarse proxy that does not test side-chain geometry or catalytic equivalence.

7. Conclusions and future work

We have described and released *Protein Function Rescue*, a reproducible, laptop-scale pipeline that converts pre-computed AlphaFold structures into ranked, evidence-linked functional hypotheses for proteins that sequence-based methods leave unannotated, together with an interactive viewer. A pilot on *M. tuberculosis* recovered three larger fold/domain hypotheses and one short-fragment false positive. The cleanest whole-chain result, P9WQ67, remains an aminotransferase-fold hypothesis rather than a reaction assignment. A controlled leave-one-out benchmark then quantified the approach on 227 enzymes across 29 families and six classes: 96.5 % exact-EC recovery, versus 93.8 % for a deliberately simple symmetric pairwise-identity baseline at the exact-reaction level; no paired significance analysis was performed. Twenty-four correct structural calls had <30 % identity to their structural hit, and the threshold gave precision 0.982 at TM ≥ 0.5.

Natural extensions include: (i) a whole-proteome run with the Foldseek back-end against the full PDB and Swiss-Prot; (ii) a larger, harder benchmark (e.g. CATH/SCOPe superfamilies, time-split characterised proteins) and comparison against profile-HMM and learned function predictors; (iii) automatic transfer and geometric verification of catalytic residues from the matched reference; (iv) incorporation of AlphaFold's predicted aligned error (PAE) for domain-aware search; and (v) prioritisation of candidates for experimental validation in collaboration with a wet laboratory.

Data and code availability

All source code, configuration, the pilot outputs (ranked candidate list as JSON/CSV, the reference-library manifest, and the functional-residue verification), and the complete validation benchmark (dataset accession list, per-query predictions, metrics, the weight-sensitivity and candidate-pool analyses, and all figure-generating code under `benchmark/`) are available in the project repository. Input structures are freely available from the AlphaFold Protein Structure Database (https://alphafold.ebi.ac.uk) and protein annotations from UniProt (https://www.uniprot.org). The reference-library and benchmark manifests list the exact UniProt accessions used, enabling exact reproduction.

Software versions: Python 3.12; AlphaFold DB model version 6; tmtools 0.3.0; gemmi ≥ 0.6; NumPy ≥ 1.24; SciPy 1.18; Biopython 1.87; Matplotlib 3.x. Default parameters: mean-pLDDT threshold 70; TM-score threshold 0.5; reference count 120; grid spacing 1.2 Å; PSP threshold 5/7; composite weights (0.5, 0.2, 0.3). Benchmark: 227 enzymes, 29 EC families, classes 1–6, length 80–520 residues.

Author contributions

[Student Name] designed the study, implemented the pipeline, performed the analysis, and wrote the manuscript. [Add supervisor/contributors as appropriate.]

Competing interests

The author declares no competing interests.

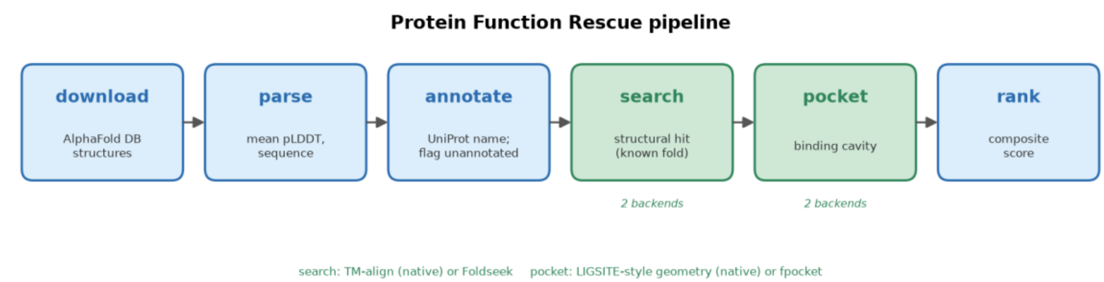
Use of generative AI

In preparing this work the author used Anthropic's Claude (Claude Opus 4 family) as a coding and writing assistant: to help design and implement the software pipeline and the validation benchmark, to generate the figures, and to draft and revise the text of this manuscript. After using this tool, the author reviewed, edited, and verified the content, and takes full responsibility for it — including all code, analyses, numerical results, and claims. No generative-AI system is listed as an author, consistent with COPE and ICMJE guidance that authorship requires accountability a non-human tool cannot hold. All citations were checked against the primary literature prior to submission.

Acknowledgements

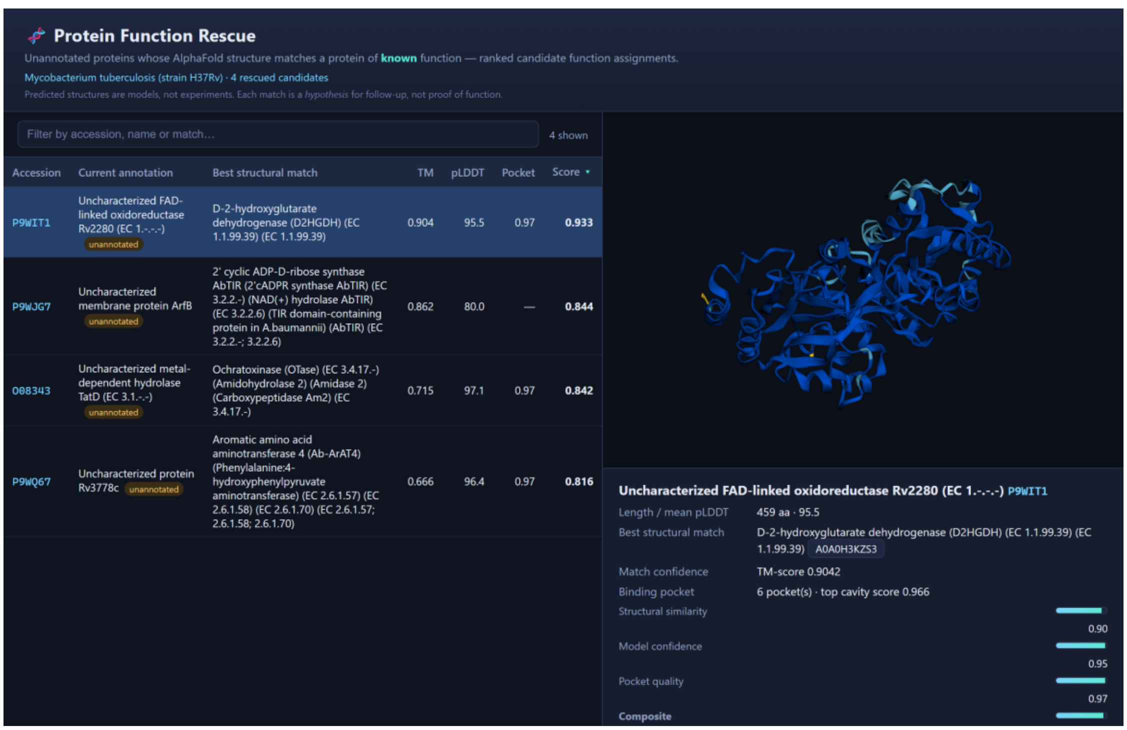
This work uses data from the AlphaFold Protein Structure Database (Google DeepMind and EMBL-EBI) and UniProt. [Add mentors, institutions, funding as appropriate.]

Figures



Pipeline overview

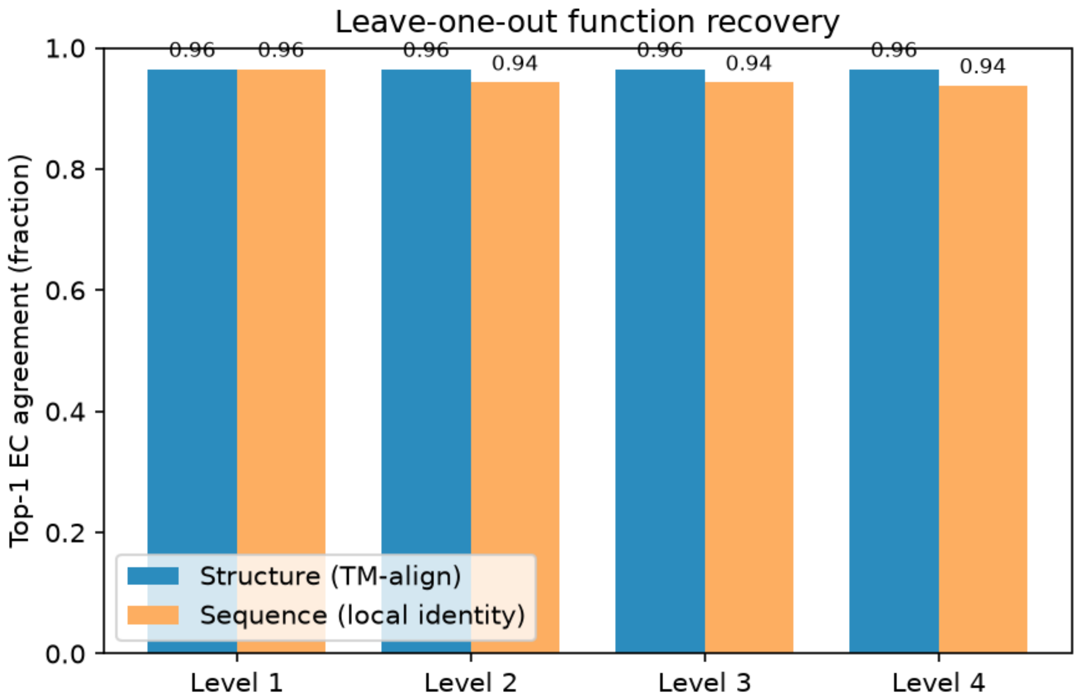
Figure 1. Pipeline overview. Six cached stages transform raw AlphaFold models into ranked candidate function assignments: download (AlphaFold DB) → parse (mean pLDDT, sequence) → annotate (UniProt; flag unannotated proteins) → search (TM-align vs. a reference library of known enzymes) → pocket (LIGSITE-style cavity detection) → rank (composite score). The search and pocket stages each have a native laptop back-end (default) and an optional high-performance back-end (Foldseek, fpocket).



Web viewer

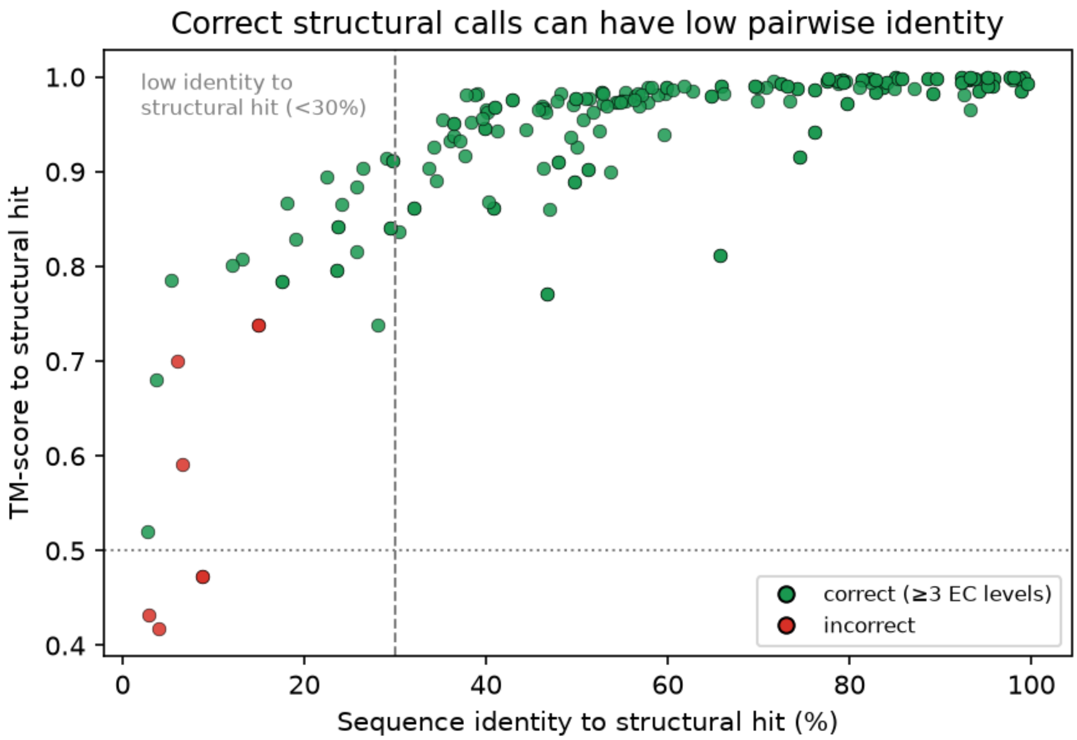
Figure 2. Web viewer. Static browser interface showing the ranked candidate table (left) and, for the selected top candidate (P9WIT1), its AlphaFold structure coloured by pLDDT (blue = high confidence) together with its best structural match (D-2-hydroxyglutarate

dehydrogenase; query/reference-normalised TM-scores 0.90/0.42), match confidence, pocket statistics, and per-component scores (right).



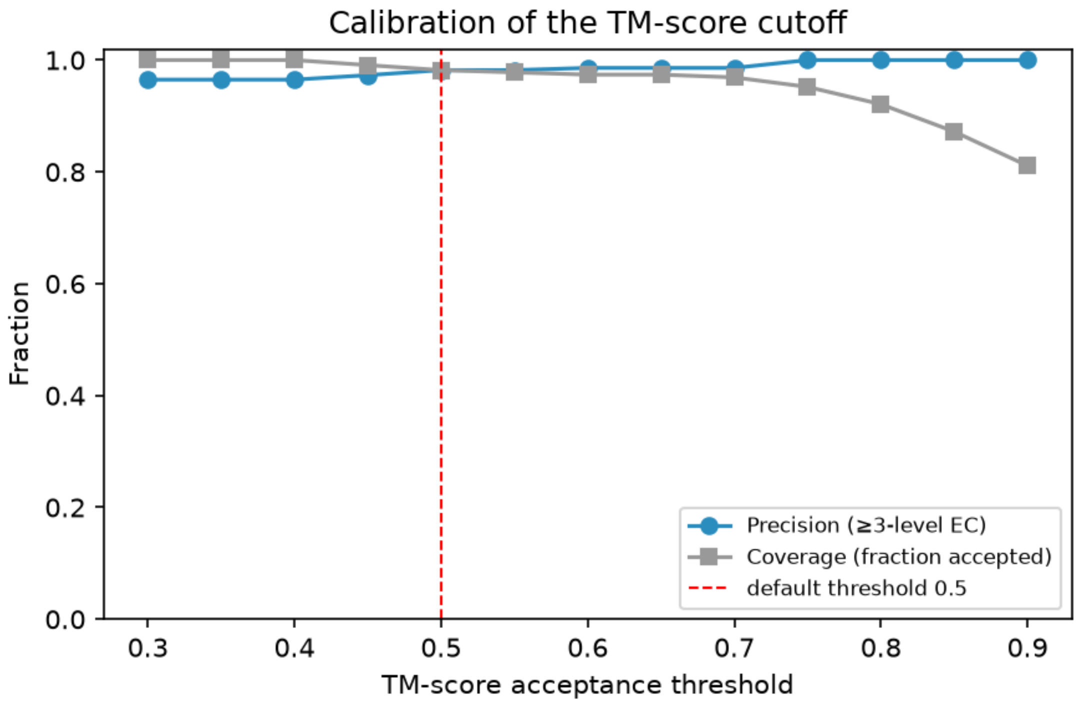
Leave-one-out function recovery

Figure 3. Leave-one-out function recovery (`benchmark/results/figures/fig_ec_accuracy.png`). Top-1 EC-number agreement at levels 1–4 for the structural method (TM-align) versus the pairwise sequence-identity baseline, over 227 enzymes in 29 EC families. Structure recovers the exact reaction (level 4) for 96.5 % of proteins, versus 93.8 % for sequence; the two methods are tied at the class level (level 1).



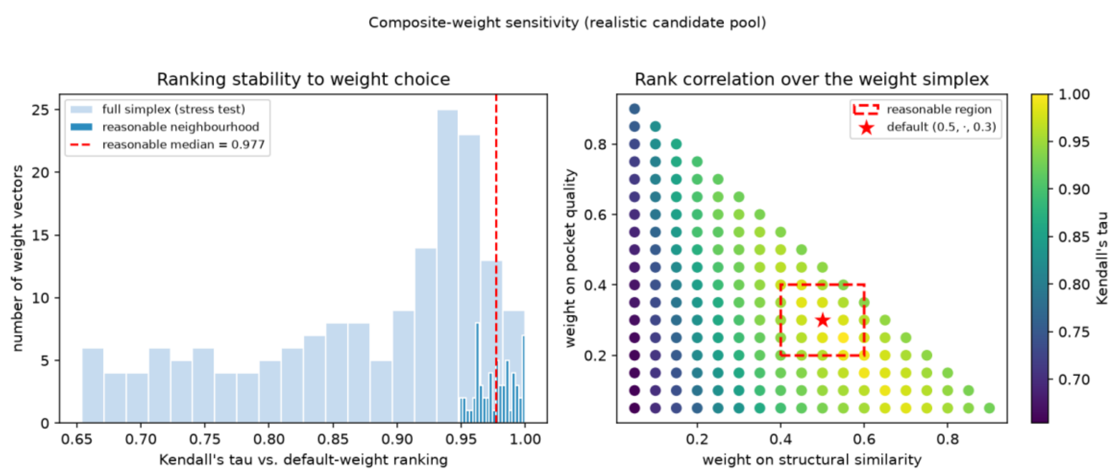
Correct structural calls can have low pairwise identity

Figure 4. Correct structural calls can have low pairwise identity (`benchmark/results/figures/fig_tm_vs_identity.png`). Each point is one protein's best structural hit, plotted as TM-score against the sequence identity to that structural hit, coloured by whether the hit's EC number is correct (≥3 levels). The dashed line marks 30 % identity; 24 correct structural calls (green) lie to its left. These are low-identity structural recoveries, not necessarily cases where the sequence baseline failed because its nearest hit may differ. The high-TM incorrect points (red) are the four fold-degeneracy false positives (e.g. the nucleoside-phosphorylase↔isomerase pair at TM ≈ 0.74).



Calibration of the TM-score threshold

Figure 5. Calibration of the TM-score threshold (`benchmark/results/figures/fig_precision_threshold.png`). Precision (fraction of accepted hits with ≥3-level EC agreement) and coverage (fraction of proteins accepted) as functions of the TM-score acceptance threshold. The default threshold of 0.5 gives precision 0.982 at 98.2 % coverage; 0.75 gives 100 % precision at 95.2 % coverage.



The candidate ranking is insensitive to the composite weights

Figure 6. The candidate ranking is insensitive to the composite weights (`benchmark/results/figures/fig_weight_sensitivity.png`). Left: distribution of

Kendall's τ between the default-weight ranking and rankings under other weight vectors, for a realistic pool of 40 rescue candidates — within a reasonable neighbourhood of the default (dark bars) τ concentrates near 1.0 (median 0.98), and even over the full simplex (light bars) it stays high (median 0.91). Right: τ across the weight simplex (structural-similarity weight vs. pocket-quality weight); the reasonable region (dashed box) around the default (star) is uniformly high.

References

1. Perdigão N, Heinrich J, Stolte C, *et al.* Unexpected features of the dark proteome. *Proceedings of the National Academy of Sciences* 2015; 112(52):15898–15903.
2. Illergård K, Ardell DH, Elofsson A. Structure is three to ten times more conserved than sequence — a study of structural response in protein cores. *Proteins* 2009; 77(3):499–508.
3. Jumper J, Evans R, Pritzel A, *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* 2021; 596:583–589.
4. Varadi M, Anyango S, Deshpande M, *et al.* AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research* 2022; 50(D1):D439–D444. (See also the 2024 database update.)
5. van Kempen M, Kim SS, Tumescheit C, *et al.* Fast and accurate protein structure search with Foldseek. *Nature Biotechnology* 2024; 42:243–246.
6. Zhang Y, Skolnick J. TM-align: a protein structure alignment algorithm based on the TM-score. *Nucleic Acids Research* 2005; 33(7):2302–2309.
7. Zhang Y, Skolnick J. Scoring function for automated assessment of protein structure template quality. *Proteins* 2004; 57(4):702–710.
8. Wojdyr M. GEMMI: A library for structural biology. *Journal of Open Source Software* 2022; 7(73):4200.
9. The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research* 2023; 51(D1):D523–D531.
10. Hendlich M, Rippmann F, Barnickel G. LIGSITE: automatic and efficient detection of potential small molecule-binding sites in proteins. *Journal of Molecular Graphics and Modelling* 1997; 15(6):359–363, 389.
11. Le Guilloux V, Schmidtke P, Tufféry P. Fpocket: an open source platform for ligand pocket detection. *BMC Bioinformatics* 2009; 10:168.
12. Gligorijević V, Renfrew PD, Kosciolek T, *et al.* Structure-based protein function prediction using graph convolutional networks. *Nature Communications* 2021; 12:3168.
13. Thornton JM, Laskowski RA, Borkakoti N. AlphaFold heralds a data-driven revolution in biology and medicine. *Nature Medicine* 2021; 27:1666–1669.

14. Akdel M, Pires DEV, Pardo EP, *et al.* A structural biology community assessment of AlphaFold2 applications. *Nature Structural & Molecular Biology* 2022; 29:1056–1067.

Note on references: citation details were checked against primary records on 13 July 2026; author lists are abbreviated with “et al.” where appropriate.

Appendix A. How to reproduce

Everything below runs on a commodity laptop (no GPU, no admin rights). Times are approximate on a consumer CPU.

Environment (~2 min).

```
git clone <repository-url> && cd protein-function-rescue
python -m pip install -r requirements.txt
```

Reproduce the pilot (Section 3) and the interactive viewer (~19 min in the verification run).

```
python -m pipeline.run all            # download → parse → annotate → search → pocket → rank
# open web/index.html in a browser (double-click) to explore the ranked candidates in 3D
```

`--limit N` restricts to the first N structures for a quick test. Outputs land in `results/` (`results.json`, `candidates.csv`) and `web/data.js`.

Reproduce the functional-residue verification (Section 3.6, Table 2, ~1 min).

```
python -m pipeline.functional_residues        # writes results/functional_residues.json
```

Reproduce the validation benchmark (Section 4, Figures 3–5; ~2–3 h, resumable).

```
python benchmark/run_benchmark.py             # builds the labelled set, runs leave-one-out
```

Writes `benchmark/results/metrics.json`, `per_query.csv`, and the figures. The O(n^2) TM-align sweep checkpoints periodically and resumes if interrupted.

Reproduce the weight-sensitivity analysis (Section 4.6, Figure 6; ~30 min).

```
python benchmark/build_candidate_pool.py      # realistic candidate pool
python benchmark/weight_sensitivity.py        # writes weight_sensitivity.json + figure
```

Render this manuscript to DOCX/PDF.

```
python paper/render.py                          # DOCX always; PDF if MS Word or a LaTeX engine is present
```

Switch to the high-performance back-ends (optional, Linux/macOS). Install Foldseek and fpocket (`docs/SETUP_TOOLS.md`), then set `structural_search.backend: foldseek` and `pocket.backend: fpocket` in `config.yaml`. **Change organism** by editing the `organism` block in `config.yaml` to any proteome listed on the AlphaFold DB FTP.

Exact input identities are fixed by the tracked manifests (`references/manifest.json`, `benchmark/results/dataset.json`), so runs are reproducible up to upstream AlphaFold DB / UniProt updates.